PCCP

PAPER

View Article Online
View Journal | View Issue

Check for updates

Cite this: *Phys. Chem. Chem. Phys.*, 2025, **27**, 3124

Ethanolamine-mediated microstructural transitions within terpenoid- and fatty acid-based deep eutectic solvents†

Anjali and Siddharth Pandey *

Deep eutectic solvents (DESs) have emerged as solubilizing media of intense interest due partly to their easily tailorable physicochemical properties. Extensive H-bonding between the constituents in a two-constituent system is the major driving force for the formation of the DES. Addition of ethanolamine (MEA), a compound having H-bonding capabilities, to the DESs composed of a terpene [menthol (Men) or thymol (Thy)] and a fatty acid [*n*-decanoic acid (DA)] results in an unprecedented increase in dynamic viscosity due to the extensive rearrangement in the H-bonding network and other interactions within the system, while the liquid mixture still behaves as a Newtonian fluid. For the non-DA DES constituted of Men and Thy, this behavior is not observed. Visual color appearance, density and electrical conductivity measurements, UV-Vis and FTIR absorbance, differential scanning calorimetry, and empirical Kamlet–Taft parameters of the MEA-added DA-based DESs reveal the microstructural changes effectively. Cybotactic regions of the fluorescent microfluidity probes [1,3-bis(1-pyrenyl)propane – an intramolecular excimer forming probe, as well as perylene and 1,6-diphenylhexatriene – well-established anisotropy probes] also manifest the unprecedented increase in the viscosity of the DA-based DES system upon MEA addition. The carboxylic acid functionality of the DA plays a crucial role in bringing microstructural changes within the system as MEA is added. Physicochemical properties of DES systems can be effectively manipulated by not only changing the constituents and their compositions, but also by judicious addition of a co-solute/co-solvent. This work offers an easy and efficient way to favorably tailor the properties of interest of these environmentally-benign media.

Received 9th October 2024,
Accepted 12th January 2025

DOI: 10.1039/d4cp03878a

rsc.li/pccp

Published on 13 January 2025. Downloaded by Westlake University on 10/30/2025 1:55:24 AM.

1. Introduction

In recent decades, efforts have been invested by the scientific community in creating more cost-effective and environmentally-benign alternatives to conventional solvents that can exhibit novel and tunable solvating capabilities.[1,2] In this regard, deep eutectic solvents (DESs), usually formed from the combination of H-bond donor (HBD) and H-bond acceptor (HBA) moieties, have garnered considerable attention.[3–7] Their physicochemical attributes have been extensively documented in numerous publications in recent times.[6–13] An important yet newer subset of the survey involves the exploration of ''binary'' mixtures, which include DESs as one of the components. Such a mixture consisting of a DES and a small functional molecule can introduce a new complexity to these media and can offer tunability to their existing physicochemical properties. However, investigations pertaining to these mixtures are rare and less-detailed than those on pure DESs. While the effect of water has been extensively studied experimentally and computationally,[14–19] the interest is now shifting towards the potential offered by a mixture of DES with small organic functionalities.

By mixing a DES with a judiciously selected molecule having appropriate functionalities, not only can the physicochemical properties of the DES be tailored effectively but media having unusual properties may be obtained through disturbing/modifying the existing interactions between the HBD and HBA within the system. In this context, the knowledge of molecular-level microstructural transition within such media becomes of utmost importance. The primary objective of this work is to afford the microstructural transition within the ''binary'' mixtures of naturally-derived hydrophobic DESs (HDESs) and a small organic molecule, ethanolamine (MEA), which is a good H-bond donor as well as acceptor. MEA also facilitates the microemulsification of fatty acid-based DESs and water systems as documented in our previous research.[20] There, the water solubility curve exhibited a Gaussian type

Department of Chemistry, Indian Institute of Technology Delhi, Hauz Khas, New Delhi–110016, India. E-mail: sipandey@chemistry.iitd.ac.in; Fax: +91-11-26581102; Tel: +91-11-26596503

† Electronic supplementary information (ESI) available. See DOI: https://doi.org/10.1039/d4cp03878a

This journal is © the Owner Societies 2025

Published on 13 January 2025. Downloaded by Westlake University on 10/30/2025 1:55:24 AM.

response instead of a sigmoidal one, which is ideally expected. It was also observed that less concentration of MEA favors water solubilization, but moderate concentrations disfavour it. We have also identified an optimum concentration of MEA, which ensures a maximum water retention by the system. This unusual behavior has prompted us to investigate the microstructural transition within these fatty acid-based DESs in the presence of MEA in a comprehensive manner.

Towards this end, we have selected three (3) DESs constituted of a saturated alkyl chain fatty acid (*n*-decanoic acid; DA) and two common terpenes (thymol; Thy and menthol; Men) in a molar ratio of 1 : 1, namely (DA : Thy), (DA : Men) and (Thy : Men). We investigated the impact of the addition of varying amounts of MEA (up to mole fraction 0.50) on their physicochemical properties. Dynamic viscosity measurements carried out initially within a temperature range 293 to 343 K show an unusual increase in viscosity upon MEA addition to DA-based DESs, whereas the viscosity change in the (Thy:Men) DES was as expected. The (DA:Thy) DES system has shown a more pronounced effect with MEA addition compared to the (DA:Men) and (Thy:Men) systems. Therefore, to obtain insight into the structural transitions, if any, within the MEA-added (DA:Thy) DES system, rheology, electrical conductivity, UV-vis as well as FTIR absorbance, and differential scanning calorimetry (DSC) of this system were investigated. Furthermore, suitable optical spectroscopic probes were employed to obtain empirical Kamlet–Taft parameters of solvent polarity and H-bonding ability, and the microviscosity of the aforementioned systems was explored using a ratiometric intramolecular excimer forming fluorescent probe along with steady-state and time-resolved fluorescence anisotropy probes. This afforded insights into the microfluidity afforded by the MEA-added DES media to solutes of importance and also revealed the correlation between the unusual increase in bulk viscosity with the probe-reported microviscosity within such interesting media.

2. Experimental

2.1. Materials

Thymol (Thy; ≥99%), L-(−) menthol (Men; ≥99%), and *n*-decanoic acid (DA; ≥98%) were purchased from Sigma-Aldrich and were stored in a desiccator. DESs were prepared by simply mixing the constituents in a 1 : 1 molar ratio followed by gentle heating and constant stirring until the mixture became clear and transparent. Ethanolamine (MEA; ≥99.5%) was purchased from Sigma-Aldrich and was stored under a highly inert atmosphere. Intramolecular excimer forming fluorescent probe 1,3-bis(1-pyrenyl)propane (BPP) was acquired from molecular probes in the highest purity. Fluorescence anisotropy probes, perylene and 1,6-diphenyl-1,3,5-hexatriene (DPH), were purchased from Sigma-Aldrich and Acros Organics, respectively. All the chemicals were used as received without any further purification and were stored in a desiccator. All the possible precautions were taken to protect the samples from moisture. While the structures of the DES constituents, Thy, Men, and DA, and the fluorescent microfluidity probes, BPP, perylene, and DPH, are presented in Fig. 1, details of the DES systems used are listed in Table S1 (ESI†).

2.2. Methods

MEA was added to freshly-prepared and rigorously dried DESs in a desired mole fraction (χ_{MEA}) varying from 0 to 0.50. The mixture was kept untouched for one hour to equilibrate at

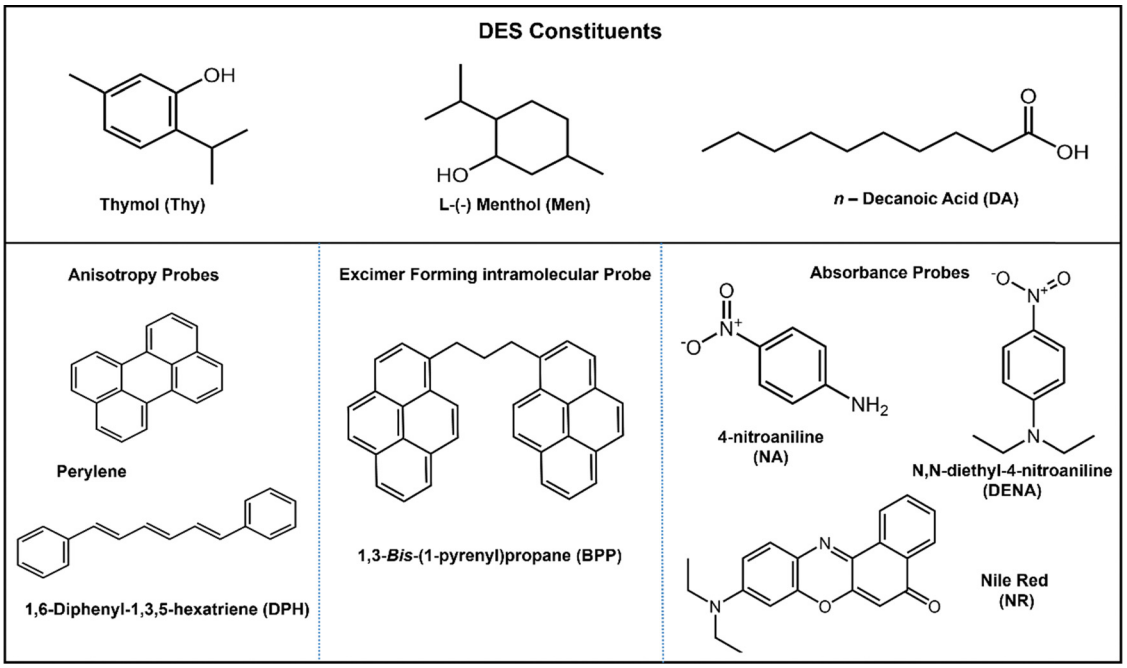


Fig. 1 Structures of the DES constituents and probes used in this work.

Published on 13 January 2025. Downloaded by Westlake University on 10/30/2025 1:55:24 AM.

room temperature in an airtight vial. The samples were then subjected to required measurements. A Mettler-Toledo, DE45 delta range density meter was used to measure the density of the samples in the temperature range of 293 to 343 K. Before carrying out the measurements, the instrument was calibrated with doubly-distilled water and air. The standard deviation estimated for the density measurement was found to be ± 0.005 g cm^{-3}. Dynamic viscosity of the same samples was measured using a Peltier-based (resolution of 0.01 K and accuracy <0.05 K) automated Anton-Paar microviscometer (model AMVn). The instrument is fitted with calibrated glass capillaries of different diameters (1.6, 1.8, 3.0, 4.0 mm) and works based on the rolling ball principle where the steel ball rolls down the inside of the inclined, sample-filled glass capillaries. The uncertainty associated with viscosity measurements was found to be 2% (± 1 relative std.). A PerkinElmer Lambda 35 double-beam spectrophotometer was used to acquire the UV-vis absorbance data at 298 K. The flow behavior of the samples was acquired on an Anton-Paar Physica MCR51 instrument having the upper plate diameter of 25 mm with a lower plate of 50 mm diameter and a gap of 0.3 mm between the plates. Measurements were performed at 298 K with a shear rate ranging from 0 to 1000 s^{-1}. The viscosity was measured by applying a constant shear rate of 50 s^{-1}. The uncertainty associated with shear rate and shear stress was found to be 3% (± 2 relative std.). The DSC thermograms of neat DESs and MEA-added DESs were obtained from 215 to 320 K on a DSC2500 instrument at a heating rate of 2 °C min^{-1}. The standard uncertainty associated with temperature and enthalpy is found to be ± 0.025 °C and ± 0.04 W g^{-1}, respectively. An inert atmosphere was maintained by purging the nitrogen gas at a flow rate of 100 mL min^{-1}. Prior to each measurement, the instrument was calibrated for temperature and heat flow each time. Infrared (IR) absorbance data was acquired from 4000 to 400 cm^{-1} on a double-beam FTIR spectrophotometer (Agilent Technologies Cary 660 ATR). Samples were well dispersed on a KBr pellet.

The 1 mM stock solution of BPP probe was prepared in ethanol; the precalculated amount of stock solution was added to a vial and ethanol was evaporated using a gentle steam of nitrogen gas. The desired amount of sample was added to achieve the final required concentration of the probe. For all the investigated samples, the probe concentration was kept constant. The same procedure was followed for the sample preparation of DPH and perylene probes for anisotropy measurements. Steady-state emission and excitation spectra of the BPP probe dissolved within the sample were acquired using a Fluorolog-3 modular spectrofluorometer purchased from Horiba-Jobin Yvon. The spectrofluorometer was equipped with excitation and emission monochromators, a 450 W Xe arc lamp as the excitation source, and a photomultiplier as a detector. Temperature was regulated using a Thermo NESLAB RTE7 circulating chiller bath, which maintained stability within ± 0.01 °C. The same instrument was used to measure the steady-state anisotropy of the perylene probe in the temperature range of 293 to 343 K. Excited-state anisotropy decay was obtained on a Horiba-Jobin Yvon, Inc. Fluorocube time-correlated single photon counting (TCSPC) fluorimeter using DPH and perylene as anisotropy probes. Perylene was excited using a Nano-LED-405 source while DPH using a Nano-LED-308. Emission was collected through the emission polarizer at a wavelength of 442 nm for perylene and 428 nm for DPH at 298 K at investigated mole fractions of MEA. Instrument response (IR) was collected using the scattering solution of glycogen in water (glycogen from bovine liver, type-IX, Aldrich). Anisotropy decays were analysed and fitted using DAS6 analysis software. All the spectroscopic measurements were repeated at least three times and the average values are reported. All the samples were prepared 2 hours in advance to achieve proper equilibration before acquiring the spectroscopic data. The blank was subtracted from each sample. Sigma Plot v15 software was utilized for the data analysis and plotting purpose.

3. Results

Depending upon the nature and the extent of interactions present, the physicochemical properties of a multicomponent mixture may deviate from their ideal additive values estimated from those of the individual components. We have investigated MEA-added DES [(Thy:DA), (Men:DA), and (Thy:Men)] systems at MEA mole fraction ranging from 0 to 0.50 considering the (Thy:Men) system as the control system. The images of the MEA-added DES (Thy:DA) system are shown in Fig. 2. It is clear from the images that MEA induces color changes in the DA-based (Thy:DA) and (Men:DA) DESs within a particular MEA concentration range. For (Thy:DA), color is induced between $0.29 \leq \chi_{MEA} \leq 0.44$, while for (Men:DA), it starts appearing at a lower MEA concentration compared to (Thy:DA), and the color is also less strong in the (Men:DA) system. However, no color is observed in the (Thy:Men) DES upon MEA addition (all the samples remain transparent throughout the entire MEA composition range from 0 to 0.50). This is the first evidence of microstructural transformation that might be taking place in DA-based DESs upon MEA addition.

3.1. Variation in density upon MEA addition

The densities of the MEA-added DES systems, [(Thy:DA)/MEA, (Men:DA)/MEA, (Thy:Men)/MEA], are measured in the temperature range of 293 to 343 K with the MEA composition ranging from 0 to 0.50 mole fraction. Fig. S1 (ESI†) shows the variations in density with the change in temperature and composition (the corresponding density values are presented in Table S2, ESI†). As expected, with increase in temperature, the density of the MEA-added DES systems decreases linearly (recovered $R^2 \geq 0.99$) due to an increase in volume for all the investigated systems. The density of the MEA-added DES systems monotonically increases as more and more MEA is added, which is expected since MEA has a higher density (ρ = 1.01 g cm^{-3} at 298 K, for example) than the density of the neat DESs [for (Thy:DA), ρ = 0.930 g cm^{-3}; for (Men:DA), ρ = 0.896 g cm^{-3}; and for (Thy:Men), ρ = 0.933 g cm^{-3} at 298 K]. It is important to note that the density of the (Thy:DA) system varies linearly with MEA

 This journal is © the Owner Societies 2025

Published on 13 January 2025. Downloaded by Westlake University on 10/30/2025 1:55:24 AM.

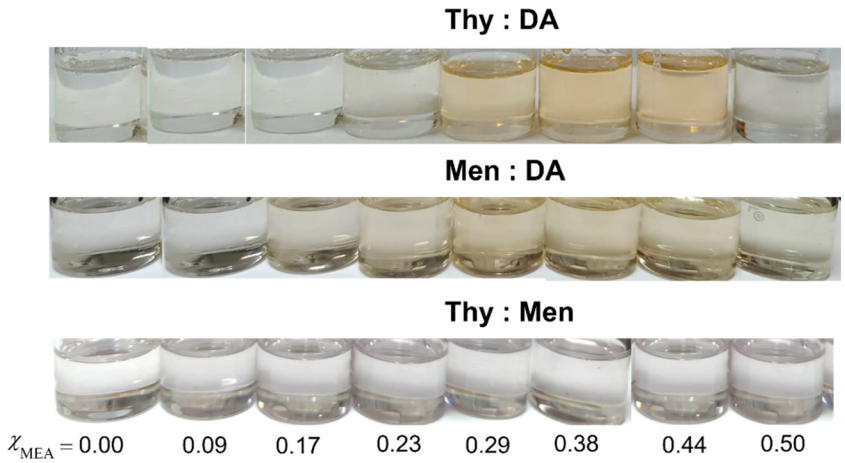


Fig. 2 Images of MEA-added DES systems at the investigated mole fractions of MEA (χ_{MEA}) under ambient conditions.

mole fraction, whereas that of the (Men:DA) and (Thy:Men) systems exhibits a quadratic dependence on MEA mole fraction.

3.2. Effect of MEA addition on dynamic viscosity

An interesting trend is observed in the dynamic viscosity (η) of the investigated DES systems as MEA is incrementally added. The experimentally-measured η of the MEA-added DES samples at 10 °C intervals at temperature ranging from 293 to 343 K and MEA composition varying from $0 \leq \chi_{MEA} \leq 0.50$ are reported in Table S3 (ESI†) (for convenience, variation in η *versus* χ_{MEA} at 293 K is presented in Fig. 3). Careful examination of Fig. 3 and entries in Table S3 (ESI†) reveals an interesting outcome. The η of MEA at 293 K is 24.08 (±0.05) mPa s. For each of the DES systems, η increases monotonically as MEA is added, with the values becoming even higher than that of neat MEA and neat DES. While the increase is gradual for the MEA-added (Thy:Men) system, surprisingly, for both DA-based DES systems, as MEA is added, the η increases drastically. For MEA-added DA-based DES systems, not only does the η become significantly higher than the η of both the neat DES [at 293 K, 15.08 and 20.41 mPa s for (Thy:DA) and (Men:DA), respectively] and neat MEA, it almost exponentially increases with an increase in MEA mole fraction (Fig. 3). This indicates that the presence of –OH and $-NH_2$ functionalities evokes strong intermolecular interactions between DES and MEA (substantially more than the intermolecular interactions present within the DES). Irrespective of the temperature, for DA-based DESs, a sudden rise in η is noticed at $\chi_{MEA} \sim 0.3$, where MEA induces yellow color in the sample (Fig. 2, *vide supra*), and afterwards, η rises drastically with further addition of MEA. The gradual increase in η of the (Thy:Men) DES system upon MEA addition is very different from that observed for DA-based DES systems. The critical role of the presence of a carboxylic acid functionality on one of the DES constituents DA in forming extensive H-bonding giving rise to anomalously high η is clearly established.

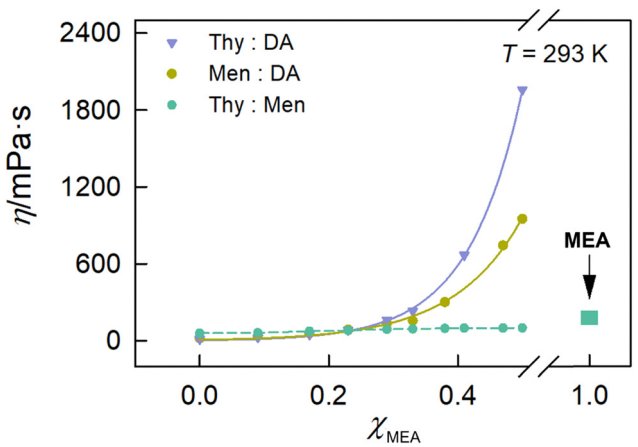


Fig. 3 Dynamic viscosities (η) of MEA-added DES systems *versus* mole fraction of MEA (χ_{MEA}) at pressure P = 101.3 kPa s and temperature T = 293 K. Solid curves represent the fit to the exponential growth expression with $R^2 > 0.99$ for MEA-added (Thy:DA) and (Men:DA) systems. The dashed curve for the MEA-added (Thy:Men) system is just to guide the eyes. The uncertainty associated with viscosity measurements is 2%.

The contrasting trend in DA-based DES systems *versus* the one where DA is not present is further highlighted through composition-dependent η data as presented in Fig. 4. $\ln \eta$ varies nonlinearly with MEA mole fraction in all the investigated systems. Notably, as expected, η does not follow the ideal-additive behavior. In the case of DA-based DESs, $\ln \eta$ *vs.* χ_{MEA} fits well to the model proposed by Gunsberg and Nishaan[21] according to the expression:

$$\ln \eta_{exp} = \chi_{DES} \ln \eta_{DES} + \chi_{MEA} \ln \eta_{MEA} + \varepsilon \chi_{MEA} \chi_{DES} \quad (1)$$

where ε is an additional empirical interaction parameter. The obtained ε and the goodness-of-fit in terms of R^2 at 298 K are tabulated in Table S4 (ESI†). The recovered R^2 further validates the fitting model for the DA-based DES systems. For the (Thy:Men) system, however, Oswal and Desai's model[22] was found to fit the composition-dependence η variation satisfactorily:

Published on 13 January 2025. Downloaded by Westlake University on 10/30/2025 1:55:24 AM.

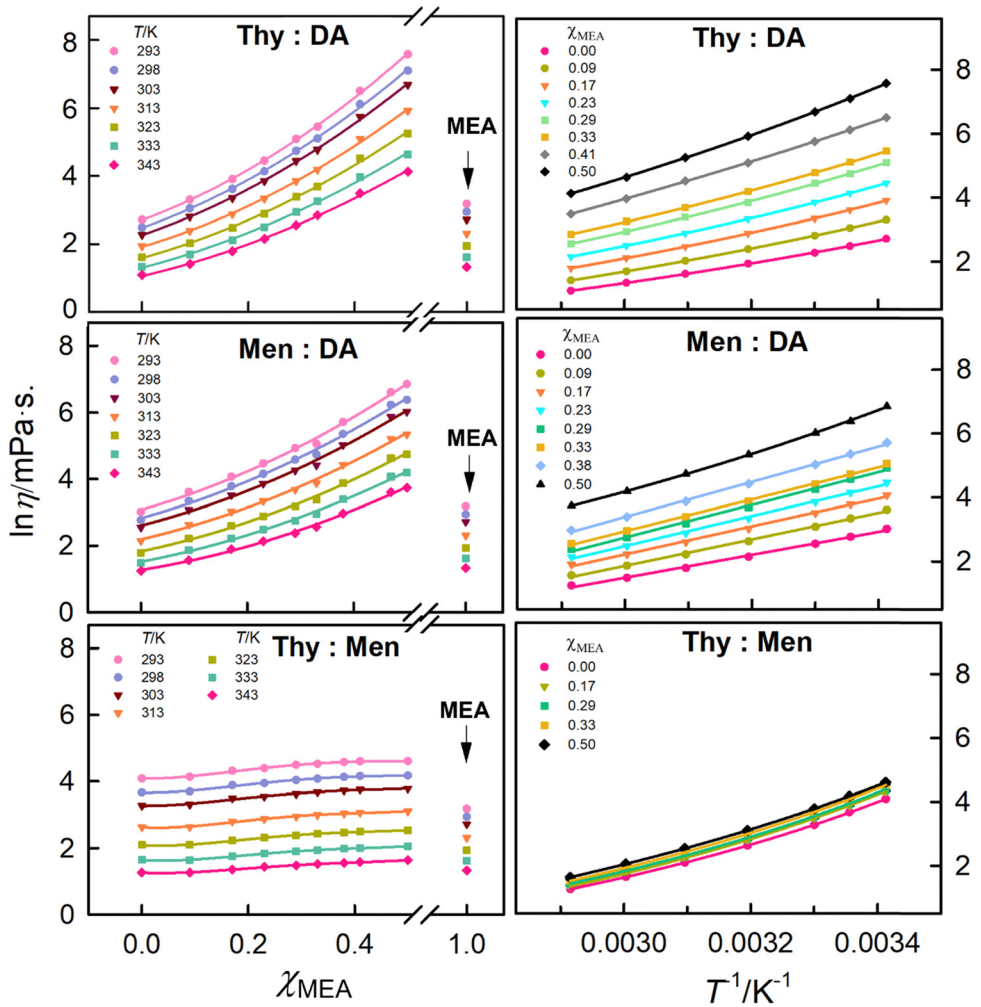


Fig. 4 Logarithm of dynamic viscosity (η) at the investigated mole fraction of MEA ($0.0 \leq \chi_{MEA} \leq 0.50$) in the temperature range of 293–343 K at pressure P = 101.3 kPa. In the mole fraction dependence of viscosity, the solid line represents fit to the Gunsberg and Nishaan model according to the expression: $\ln \eta_{exp} = \chi_1 \ln \eta_1 + \chi_2 \ln \eta_2 + \varepsilon \chi_1 \chi_2$ for the (Thy:DA) DES system and (Men:DA) DES system while for the (Thy:Men) DES, fits are for the expression: $\ln \eta_{exp} = \chi_1 \ln \eta_1 + \chi_2 \ln \eta_2 + \varepsilon \chi_1 \chi_2 + K \chi_1 \chi_2 (\chi_1 - \chi_2)$[1:DES; 2:MEA]. However, in the temperature dependence of viscosity, the solid line represents fit to the VFT equation: $\ln \eta = A + \frac{B}{T - T_0}$. The uncertainty associated with viscosity measurements is 2%.

$$\ln \eta_{exp} = \chi_{DES} \ln \eta_{DES} + \chi_{MEA} \ln \eta_{MEA} + \varepsilon \chi_{MEA} \chi_{DES} + K \chi_{MEA} \chi_{DES} (\chi_{DES} - \chi_{MEA}) \quad (2)$$

where an additional parameter K needed to be introduced. Table S5 (ESI†) reports the recovered empirical parameters obtained from the fitting of the η data to eqn (2) for the MEA-added (Thy:Men) system. Notably, the value of the interaction parameter, ε, is observed to be negative for MEA-added DA-based DES systems, whereas it is positive for the MEA-added (Thy:Men) system. Previous studies claim that a negative value of ε indicates positive deviation from Raoult's law, whereas a positive value of ε indicates negative deviation. In this context, (Thy:DA) exhibits more positive deviation than (Men:DA), while (Thy:Men) exhibits a negative deviation. Interestingly, the recovered ε values also show an almost systematic variation with temperature. In DA-based DESs, with increasing temperature, deviation from Raoult's law decreases, and thereby, the system is expected to show more ideal mixing behavior at higher temperatures. While in (Thy:Men), deviation from Raoult's law becomes more pronounced at higher temperatures; hence, at lower temperatures, the system is expected to exhibit more ideal mixing behavior. Furthermore, to elucidate the intermolecular interaction within MEA-added DES systems, the excess logarithmic viscosity, $(\ln \eta)^E$, is calculated according to the following expression:[23]

$$(\ln \eta)^E = \ln \eta - [\chi_{MEA} (\ln \eta_{MEA}) + \chi_{DES} (\ln \eta_{DES})] \quad (3)$$

and plotted in Fig. 5. A detailed examination of Fig. 5 reveals that $(\ln \eta)^E$ is positive (synergistic effect) for all the investigated MEA compositions and temperatures. Additionally, $(\ln \eta)^E$ consistently decreases with increasing temperature across all the systems, which potentially suggests that H-bonding plays a critical role in the interaction between DES and MEA. More importantly though, the magnitude of $(\ln \eta)^E$ is significantly higher for DA-constituted DESs as opposed to the (Men:Thy)

 This journal is © the Owner Societies 2025

Published on 13 January 2025. Downloaded by Westlake University on 10/30/2025 1:55:24 AM.

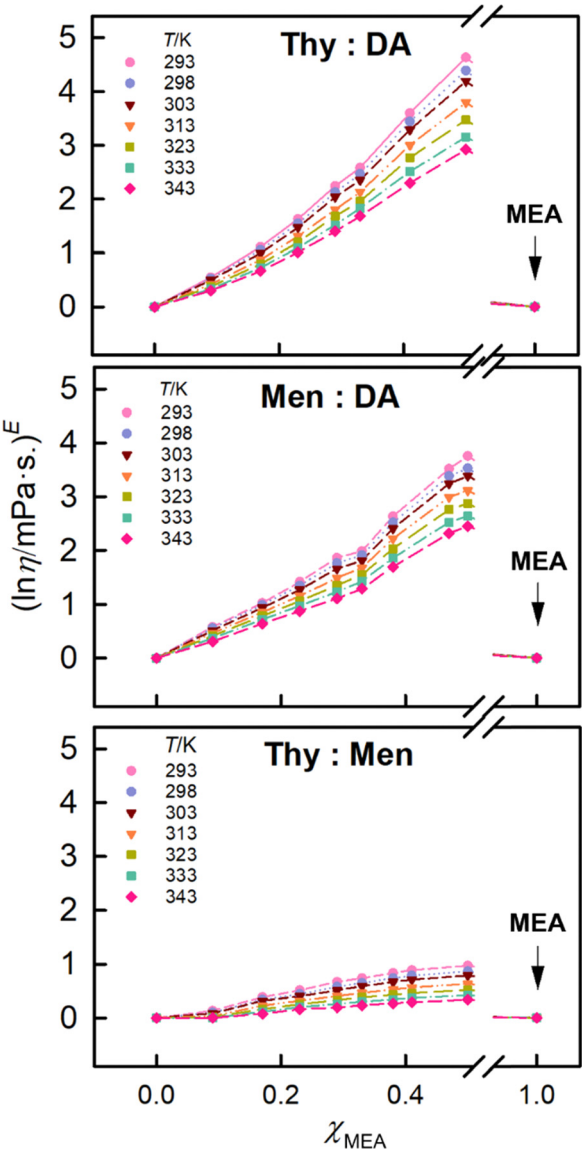


Fig. 5 Variations in $(\ln\eta)^E$ with mole fraction of MEA (χ_{MEA}) as temperature is increased from 293 to 343 K for the investigated DES systems. Dotted curves are just to guide the eyes. The uncertainty associated with viscosity measurements is 2%.

DES, indicating that, along with H-bonding, stronger attractive interactions likely exist between DA-based DESs and the –OH/–NH_2 functionalities of MEA that can contribute to the pronounced rise in viscosity observed for DA-based DESs upon MEA addition. The clear differences originating from the presence of DA as a constituent of the DES are further highlighted.

3.3. Temperature dependence of dynamic viscosity

As expected, dynamic viscosity decreases monotonically with increasing temperature for each system (Table S3, ESI† and Fig. 4). The Vogel–Fulcher–Tammann (VFT) model[24] has shown better correlation between η and T in the temperature range of 293 to 343 K rather than the Arrhenius model:

$$\ln\eta = A + \frac{B}{T - T_0} \quad (4)$$

where A and B are empirical parameters, and T_0 is the fixed temperature where the configurational entropy of the system completely vanishes. Temperature dependence of the viscosity is fitted to the VFT equation and the recovered parameters are reported in Table S6 (ESI†) along with the goodness-of-fit in terms of R^2. The R^2 further validates the good adherence to the VFT model for all the systems investigated here (plot of ln η *vs.* T^{-1} is shown in Fig. 4 along with the best fit curves). The activation energy of the viscous flow ($E_{a,\eta}$) at 298 K is calculated from the recovered parameters according to the expression:

$$E_{a,\eta} = RB\left(\frac{T}{T - T_0}\right)^2 \quad (5)$$

It is interestingly to note the trends in $E_{a,\eta}$ among the MEA-added DES systems (Table S6, ESI†). The $E_{a,\eta}$ of neat MEA is 34.3 (±1.8) kJ mol^{-1}. While the $E_{a,\eta}$ values of (Thy:DA) and (Men:DA) are fairly similar in magnitude [31.1 (±1.8) and 35.5 (±2.0) kJ mol^{-1}, respectively], for (Thy:Men) the $E_{a,\eta}$ is much higher [60.2 (±2.9) kJ mol^{-1}] suggesting interactions to be different when MEA is added to this DES as opposed to DA-based DESs. This further corroborates the MEA composition dependent η data discussed in the previous section. Interestingly, addition of MEA does not result in any significant changes in the $E_{a,\eta}$ of the (Thy:Men) system, whereas for DA-based DESs, the $E_{a,\eta}$ increases significantly [to 66.2 and 55.5 kJ mol^{-1} for (Thy:DA) and (Men:DA), respectively, for 0.5 mole fraction MEA addition]. Clearly, addition of MEA brings considerable changes within the DA-based DES systems as opposed to (Thy:Men) DES – the presence of a carboxylic acid group on a DES constituent plays a critical role (*vide supra*). For DA-based DESs, an almost linear increase in $E_{a,\eta}$ with increase in MEA mole fraction hints at the presence of stronger intermolecular H-bonding interactions between the DES and MEA as compared to interactions within the DES and within MEA, respectively.

3.4. Change in electrical conductivity

The electrical conductivity of the MEA-added DES systems was measured at 298 K and the results are depicted in Fig. 6. It is observed that variation of electrical conductivity with MEA addition is similar to that of dynamic viscosity in DA-based DES systems. A dramatic rise in electrical conductivity is noted at $\chi_{MEA} \sim 0.29$ in DA-based DES systems, where a corresponding rise in dynamic viscosity also occurs (*vide supra*). At $\chi_{MEA} \sim 0.50$, the electrical conductivity reaches its highest value, which is considerably higher than the value in neat (Thy:DA) DES ($\kappa \sim 1.0$ μS cm^{-1} under ambient conditions) as well as in neat MEA ($\kappa \sim 12$ μS cm^{-1} under ambient conditions).[25]

This unusual increase in electrical conductivity with addition of MEA hints at the formation of ion pairs leading to induced ionic interactions resulting in a drastic increase in

Published on 13 January 2025. Downloaded by Westlake University on 10/30/2025 1:55:24 AM.

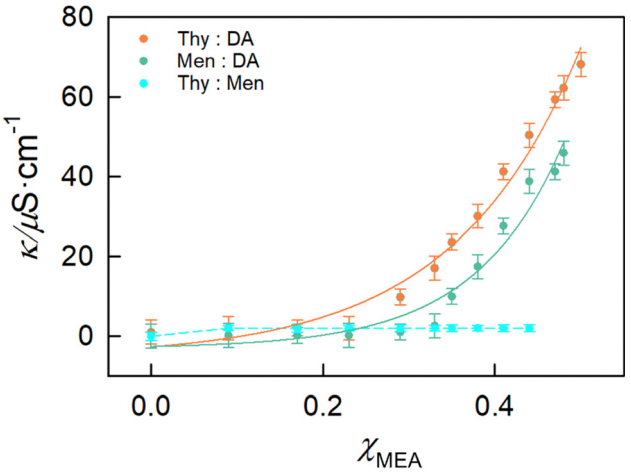


Fig. 6 Electrical conductivity (κ) variation within MEA-added DES systems at investigated mole fractions of MEA (χ_{MEA}) at 298 K. Solid curves represent fit to an exponential growth expression with $R^2 > 0.97$ for the MEA-added (Thy:DA) and (Men:DA) systems. The dashed curve for the MEA-added (Thy:Men) system is just to guide the eyes.

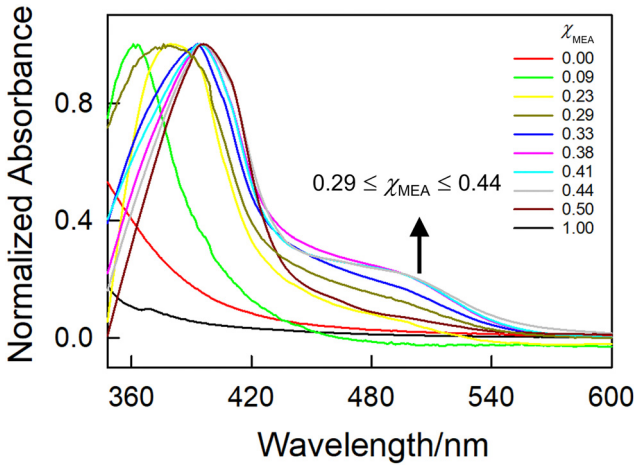


Fig. 7 UV-Vis absorbance spectra of the MEA-added (Thy:DA) DES system at the investigated mole fractions of MEA (χ_{MEA}) at 298 K.

dynamic viscosity. However, based on electrical conductivity data alone, it is difficult to predict the exact nature of such interactions. Nevertheless, we hypothesize that the interaction of the carboxylic acid group of DA with MEA results in the generation of H^+ ions. It is well established that H^+ ions have surprisingly high mobility even in viscous media due to the hopping mechanism within the water or H-bonding systems. Therefore, beyond the breakpoint in the electrical conductivity, we tentatively attribute this to the presence of H^+ ions within these media. It is interesting to note that the (Thy:Men) system does not exhibit any such increase in dynamic viscosity or electrical conductivity. In this respect, the (Thy:Men) system serves as an effective control system. This clearly establishes that the microstructural changes in MEA-added DA-based DES systems have their origin in the interactions present between the carboxylic acid group of the DA and MEA.

3.5. UV-Vis absorbance of the MEA-added (Thy:DA) DES systems

Although both MEA and the investigated DESs are clear and colorless in appearance, addition of MEA to the DES induces color in the DA-based DES systems, while no color appeared in the MEA-added (Thy:Men) DES system (*vide supra*). Fig. 7 presents UV-vis molecular absorbance spectra of the (Thy:DA) DES system, which shows the emergence of a new band in the 330–420 nm range as MEA is added. Notably, as the amount of MEA increases, the band maxima show a bathochromic shift with the emergence of a shoulder at lower energy (470–520 nm). The relative absorbance of the shoulder becomes more pronounced as MEA in the composition range 0.29–0.44 is added. Expectedly, no absorbance feature is seen for either the (Thy:DA) DES or MEA in the region >360 nm; the appearance of a prominent absorbance band upon MEA addition to the (Thy:DA) DES hints at the formation of a H-bonded network within the system perhaps giving rise to large aggregated assemblies. The emergence of the shoulder may imply morphological transformation of the aggregates. We can propose that at $\chi_{MEA} \sim 0.29$, a transformation in the aggregation process is initiated within the (Thy:DA) DES system, which results in yellow coloration due to scattering. On further addition of MEA ($0.29 \leq \chi_{MEA} \leq 0.44$), the shoulder grows as this scattering intensifies perhaps implying the growth of the aggregates, which further results in more pronounced yellow color. When χ_{MEA} exceeds 0.44, the shoulder disappears and so does the scattering, which is suggestive of the breakdown of aggregates and hence, discoloration of the sample is the result.

3.6. Rheology measurements

Rheological measurements are used next to obtain further insight into the sudden increase in the dynamic viscosity of the DA-based DESs upon addition of MEA. Our proposition of aggregation within the MEA-added (Thy/Men:DA) DES systems with χ_{MEA} ranging between 0.29 and 0.44 based on our observation from UV-vis absorbance data and the appearance of yellow color after adding MEA to DES, prompted us to conduct a rheology study of the aforementioned samples to determine whether MEA-added samples are Newtonian or non-Newtonian in nature. Fig. 8 shows the plot of shear stress *versus* shear rate for the MEA-added (Thy:DA) DES system at 298 K. All the investigated samples have shown a linear relationship between shear stress and shear rate with the goodness-of-fit $R^2 \geq 0.997$. Furthermore, the dynamic viscosity of the samples also remains largely unaffected by the variations in shear stress (Fig. S2, ESI†). Consequently, all the samples fulfill the criteria of Newtonian fluid across the χ_{MEA} ranging from 0 to 0.50, where shear stress and shear rate show a linear relationship and the dynamic viscosity also remains constant regardless of shear rate alternations unless subjected to variations in temperature. This evidently establishes that the flow behavior of all the investigated systems is Newtonian in nature. Interestingly, the respective slope of the stress-shear rate line also increases

 This journal is © the Owner Societies 2025

Published on 13 January 2025. Downloaded by Westlake University on 10/30/2025 1:55:24 AM.

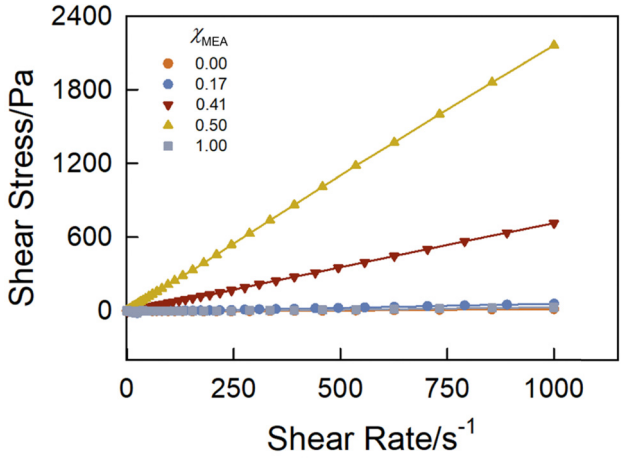


Fig. 8 Shear stress *versus* shear rate for the MEA-added (Thy:DA) DES system at the investigated mole fractions of MEA (χ_{MEA}) at temperature T = 298 K. Solid lines represent the outcomes of the linear regression analysis.

with increase in MEA proportion in the mixture, which indicates that the mixture is becoming more incompressible with the addition of MEA until the MEA mole fraction reaches 0.50.

3.7. Differential scanning calorimetry data

DSC was performed at temperature range 215 to 320 K with a heating rate of 2 °C min^{-1} [the thermal behavior of the investigated MEA-added (Thy:DA) DES system is shown in Fig. 9]. At lower temperatures, for the neat DES, we have encountered a lone endothermic event around 284 K, which is associated with the melting point of the DES as per the literature.[26] Slight discrepancies in DSC data can arise due to the mode of DES preparation, moisture content, and sensitivity of the instrument. Interestingly, with the addition of MEA, we observe qualitative changes in the endothermicity of the DES melting event. However, along with this peak, we observe another new endothermic event near the vicinity of the melting point of the (Thy:DA) DES. This event occurs at a lower temperature (*i.e.*, 270–279 K) than the (Thy:DA) DES melting temperature. It may be noted here that pure MEA melts at 279 K; therefore, the event at lower temperature (*i.e.*, 270–279 K) in the DES–MEA mixture could be associated with the melting of pure MEA. However, there are also occurrences of strong exothermic events in the presence of MEA at even lower temperature, which are absent in the neat MEA and DES, providing evidence of strong chemical interactions between the DES and MEA.

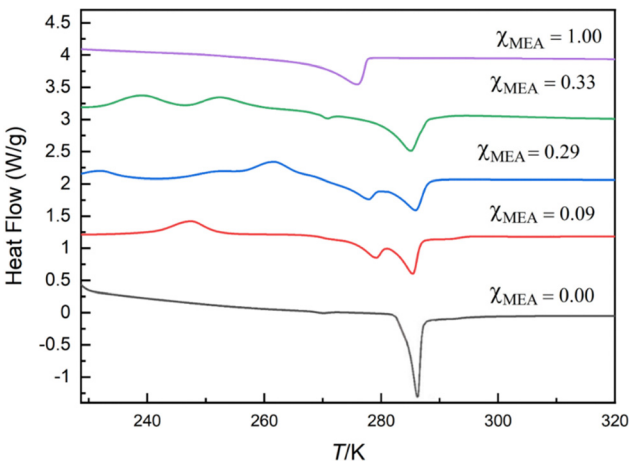


Fig. 9 DSC profile of the MEA-added (Thy:DA) system at the investigated mole fractions of MEA (χ_{MEA}) measured at a heating rate of 2 °C min^{-1}.

3.8. Changes in FTIR absorbance spectra

FTIR absorbance spectra of the MEA-added (Thy:Men) and (Thy:DA) systems are reported in Fig. 10 and 11, respectively. A detailed analysis reveals that the bands at 1232, 1156, and 1091 cm^{-1} correspond to the stretching vibrations of the phenolic C–O group, while those at 1621 and 1583 cm^{-1} are attributed to the aromatic C=C stretching vibrations.[27,28] Peaks observed in the vicinity of 2925–2872 cm^{-1} and 1458–1380 cm^{-1} correspond to C–H stretching and bending modes, respectively.[27,28] The broad band in the range of 3400–3200 cm^{-1} is attributed to the stretching frequency of the –OH bond. Upon addition of MEA to the (Thy:Men) DES system, the H-bonding profile of the DES alters very little (few bands exhibit slight shifts). The spectra of the MEA-added (Thy:Men) DES, in general, resemble that of the neat DES. This implies that there is no appreciable chemical interaction present between the (Thy:Men) DES and MEA except for slight alterations in H-bonding profile. The peak corresponding to the C=C shifts to lower wavenumbers, whereas that associated with C–O stretching vibration (1232 cm^{-1} for thymol and 1021 cm^{-1} for menthol[29]) shifts to higher wavenumbers with subsequent addition of MEA. This suggests a slight decrease in O–H stretching frequency, and hence, a slight increase in H-bonding. Probably, this extended H-bonding may be the reason for the synergistic viscosity observed in the (Thy:Men) DES

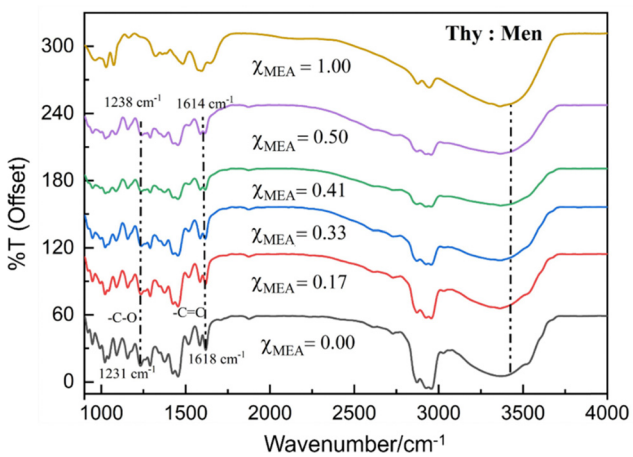


Fig. 10 FTIR absorbance spectra of the MEA-added (Thy:Men) system at the investigated mole fractions of MEA (χ_{MEA}). Dotted lines are just to guide the eyes.

Published on 13 January 2025. Downloaded by Westlake University on 10/30/2025 1:55:24 AM.

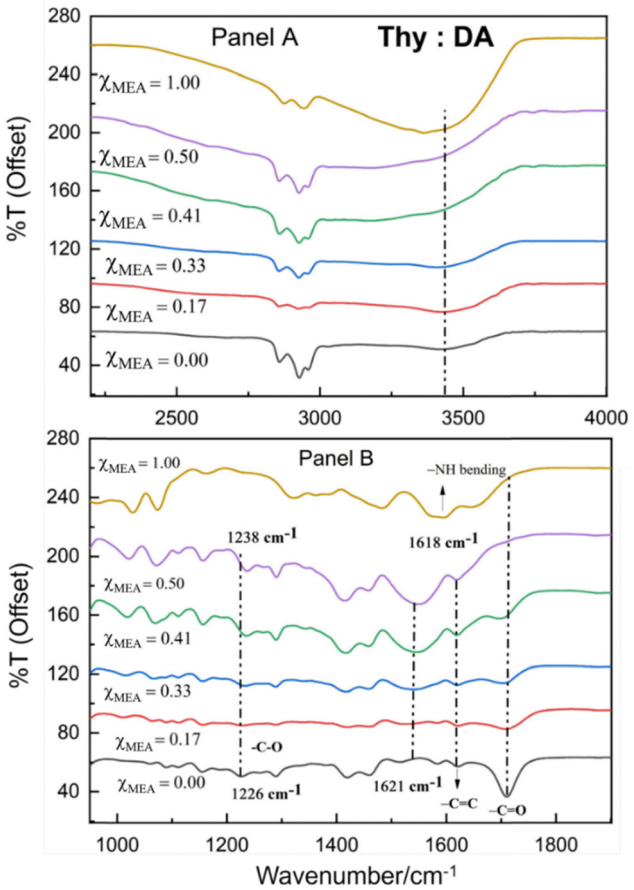


Fig. 11 FTIR absorbance spectra of the MEA-added (Thy:DA) system at the investigated mole fractions of MEA (χ_{MEA}). Dotted lines are just to guide the eyes.

system, although to a much lesser extent as compared to DA-based MEA-added DES systems.

However, beyond χ_{MEA} = 0.33, the shifts in the IR peaks become more prominent – the spectra no longer resemble the spectrum of neat DES or neat MEA. The H-bonding envelope also broadens and shifts towards a lower wavenumber [from ~(3400–3600 cm^{-1}) to ~(3100–3600 cm^{-1})] which strongly suggests an extension of H-bonding resulting from DES and MEA interactions. Additionally, the spectral peak corresponding to N–H bending of MEA (1595 cm^{-1}) does not appear even when 0.50 mole fraction of DES is added to the (Thy:DA) DES, which implies the $–NH_2$ functionality of MEA within the system to be not free; it is occupied in forming H-bonding.[30] The addition of MEA also results in the emergence of a new peak in the spectral region of 1520–1550 cm^{-1}, whereas the intensity of the carbonyl peak at ~1710 cm^{-1} decreases and completely vanishes at χ_{MEA} = 0.50. This indicates that the carbonyl group of DA engages in chemical interaction with MEA and establishes new interactions by forming new fluxional bonds.

In contrast, the spectral features of thymol do not show any statistically meaningful changes. Overall, in the (Thy:DA) system, it appears that not only H-bonding but certain chemical interactions between DA and MEA also contribute to the observed synergism observed in dynamic viscosity (*vide supra*). Fig. 11 (panels A and B) illustrates the FTIR absorbance spectra of the MEA-added (Thy:DA) system. A careful examination reveals that the stretching frequency of the phenolic C–O group (1226 cm^{-1}) significantly increases with continuous addition of MEA (by ~12 cm^{-1} for 0.50 mole fraction MEA addition), while the C=C stretching frequency decreases. This suggests the existence of conjugation in the thymol structure, and hence, the extension of the H-bonding network. In the (Thy:DA) system, it appears that up to MEA mole fraction of 0.33, there is no significant change in the H-bonding profile and the spectra closely resemble that of the neat (Thy:DA) DES.

3.9. Empirical Kamlet–Taft parameters

Empirical Kamlet–Taft (KT) parameters are widely used to establish the physicochemical properties of any solubilizing milieu. Specifically, KT parameters can provide an estimate of dipolarity/polarizability (π^*) and H-bonding ability of any complex mixture in terms of H-bond acceptor basicity (HBA, β) and H-bond donor acidity (HBD, α), respectively. UV-Vis absorbance from the solvatochromic probes – *N*,*N*-diethyl-4-nitroaniline (DENA),[31] Nile red (NR),[32] and nitroaniline (NA)[33] – are recorded to estimate π^*, α, and β, respectively, of the MEA-added (Thy:DA) DES systems under ambient conditions, according to the following expressions:

$$\pi^* = 8.649 - 0.314\bar{\upsilon}_{DENA} \quad (6)$$

$$\alpha = \frac{20.47 - 1.95\pi^* - \bar{\upsilon}_{NR}}{1.01} \quad (7)$$

$$\beta = -0.357\bar{\upsilon}_{NA} - 1.176\pi^* + 11.12 \quad (8)$$

where $\bar{\upsilon}_{DENA}$, $\bar{\upsilon}_{NR}$, and $\bar{\upsilon}_{NA}$ correspond to the lowest energy absorbance band maxima of DENA, NR, and NA, respectively, in kK units (1000 cm^{-1} = 1 kK). All the probes exhibit bathochromic shift with addition of MEA in the (Thy:DA) DES (Fig. 12). The estimated KT parameters are reported in Table S7 (ESI†).

The π^* of the neat (Thy:DA) DES is 0.86 ± 0.05 and that of neat MEA is 1.01 ± 0.05. While the π^* of the DES increases as MEA is added, interestingly, it attains values even higher than that in neat MEA (it reaches 1.10 for $\chi_{MEA} > 0.33$) indicating that the system surpasses the polarizability of the neat DES and neat MEA.

This clearly implies much higher polarity/polarizability associated with the MEA-added DA-based DESs. Moreover, before MEA addition, DES is predominately a strong HBD solvent; however, with the addition of MEA, the HBD acidity decreases and becomes even lower than that of MEA [α of neat MEA is reported to be 0.78].[34] Similarly, the HBA basicity of the MEA-added (Thy:DA) DES system is lower than that expected from ideal-additive behavior. Apparently, addition of MEA to the DES disrupts the H-bonding network within the DES and significantly reorganizes and apparently forms a newer and stronger H-bonding network within the system, causing the HBD acidity as well as HBA basicity of the medium to become lower than

 This journal is © the Owner Societies 2025

Published on 13 January 2025. Downloaded by Westlake University on 10/30/2025 1:55:24 AM.

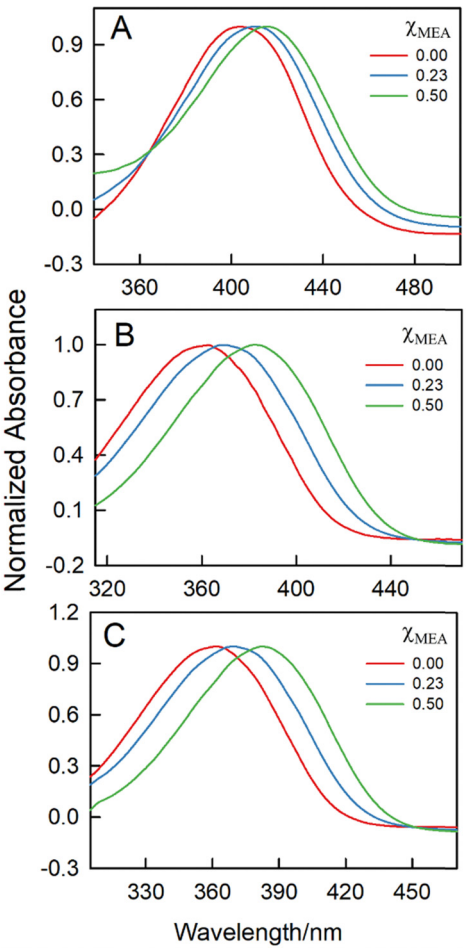


Fig. 12 Absorbance spectra of DENA (panel A), NA (panel B) and NR (panel C) dissolved in the MEA-added (Thy:DA) DES system at the investigated MEA mole fractions (χ_{MEA}) at 298 K.

that expected. The presence of strong interactions within the system is established nonetheless.

3.10. Response of the intramolecular excimer forming fluorescent microviscosity probe

Microviscosity is the viscosity/fluidity sensed by the solute molecules in their cybotactic region. The microviscosity reported by a probe may be different from the bulk dynamic viscosity depending on the presence of probe-milieu interactions within the system.[35] In order to assess whether the dramatic increase in bulk dynamic viscosity upon MEA addition to DA-based DESs is manifested through the response of probe-reported microviscosity, 1,3-bis-(1-pyrenyl)propane (BPP), a commonly used fluorescent probe, is utilized. It is well-established that BPP exhibits structured monomer bands characterizing fluorescence emission from the pyrenyl group along with a broad and structureless band with the maximum intensity lying between 450–500 nm due to the formation of an intramolecular excimer as a result of folding of the two pyrenyl moieties that depends on the viscosity of the immediate milieu surrounding BPP. As the microviscosity of the cybotactic region

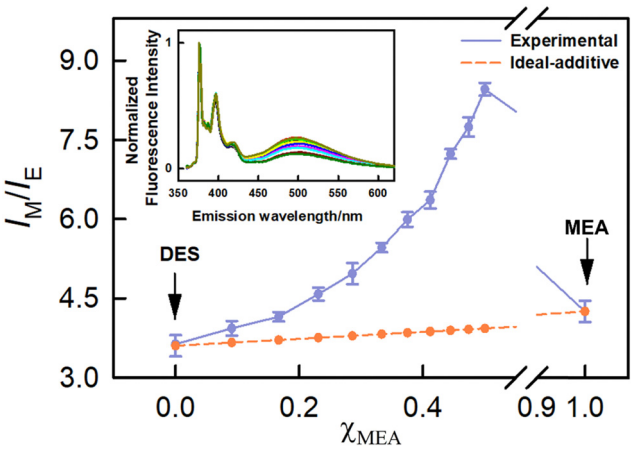


Fig. 13 Variation of experimental and ideal-additive I_M/I_E for BPP dissolved in the MEA-added (Thy:DA) DES system at the investigated mole fraction of MEA (χ_{MEA}) at 298 K. Inset shows normalized fluorescence spectra. Dotted and solid curves are just to guide the eyes.

of BPP increases, two pyrenyl moieties feel increased hindrance in folding, and hence, the fluorescence intensity of the excimer band decreases.[36] The fluorescence emission spectra of BPP dissolved in neat (Thy:DA) DES, MEA-added (Thy:DA) DES and neat MEA are acquired at 298 K (Fig. S3, ESI†). The intramolecular excimer formation is clearly depicted. The monomer (at 378 nm)-to-excimer (at 450 nm) emission intensity ratio (I_M/I_E) depicts the hindrance encountered during the folding by the two pyrenyl moieties on BPP – the higher the $\left(\frac{I_M}{I_E}\right)$, the more the cybotactic region viscosity. A normalized fluorescence spectrum of the BPP probe and the plot of $\left(\frac{I_M}{I_E}\right)$ *versus* MEA mole fraction (Fig. 13) clearly shows a dramatic increase in (I_M/I_E) with increasing χ_{MEA}. To corroborate the unusual bulk dynamic viscosity data, it is important to note that the (I_M/I_E) in the MEA-added (Thy:DA) DES system, for the most part, is higher than its value in neat MEA. To further highlight this unusual trend, ideal-additive, $(I_M/I_E)^{Ideal\text{-}additive}$, values are estimated (and presented in Fig. 13) from the expression proposed by Acree and coworkers[37] assuming the mole fraction of MEA in the cybotactic region of the BPP probe to be equal to the mole fraction of MEA in the bulk liquid:

$$\left(\frac{I_M}{I_E}\right)^{Ideal\text{-}additive} = \frac{\chi_{DES} I_{M,DES} + \chi_{MEA} I_{M,MEA}}{\chi_{DES} I_{E,DES} + \chi_{MEA} I_{E,MEA}} \quad (9)$$

It is noteworthy that the experimental (I_M/I_E) values are significantly higher than the $(I_M/I_E)^{Ideal\text{-}additive}$. Both (I_M/I_E) *vs.* χ_{MEA} and η *vs.* χ_{MEA} trace similar paths, which confirms that bulk viscosity correlates well with the microviscosity of the probe BPP. This evidently suggests that dynamic viscosity is molecular interaction-driven (reflected by microviscosity) and these interactions become stronger as the MEA proportion increases within the DES. The microenvironment of the probe

Published on 13 January 2025. Downloaded by Westlake University on 10/30/2025 1:55:24 AM.

BPP cybotactic region effectively mimics the bulk solubilization milieu.

Fluorescence excitation scans of BPP are acquired next in all the investigated systems (varying MEA composition in DES) at monomer (λ_{em} = 376 nm) and excimer (λ_{em} = 496 nm) emission wavelengths, respectively, at 298 K (Fig. S4, ESI†). A lower concentration of probe is considered to prevent intermolecular aggregation. Fig. S4 (ESI†) reveals interesting outcomes. Excitation scans in neat (Thy:DA) DES and neat MEA at the two wavelengths overlap perfectly. However, in the MEA-added (Thy:DA) DES system, it is not the case. While the overlap of excitation scans collected at monomer and excimer emission confirms excimer formation to be exclusively taking place in the excited-state (no ground-state heterogeneity), the lack of overlap indicates the presence of ground-state heterogeneity in the solubilizing media. For the MEA-added (Thy:DA) DES systems, ground-state heterogeneity is revealed, which hints at cyclization followed by dimer formation in the ground-state. We believe that the highly polar H-bonded network within the MEA-added (Thy:DA) DES affords poor solvation to highly nonpolar BPP, thus leading to the presence of some population of the folded form in the ground-state.

Furthermore, at χ_{MEA} = 0.50, the MEA-added (Thy:DA) DES exhibits ~40 times higher dynamic viscosity (~1900 mPa s) than those reported for systems, such as, ionic liquid 1-butyl-3-methylimidazolium triflate ([bmim][Tf_2N] with η = 44.4 mPa s) and tetraethylene glycol (TEG with η = 46.1 mPa s).[35] In spite of this, BPP (I_M/I_E) in the 0.50 mole fraction MEA-added (Thy:DA) system is lower (I_M/I_E = 8.5) than the I_M/I_E in [bmim][Tf_2N] (I_M/I_E = 9.7) and that in TEG (I_M/I_E =16.1). This prompted us to further investigate the scaling of microviscosity to assess whether it is due to any specific interactions of the probe with the above reported systems as the microviscosity of the MEA-added DES system is observed to be less in comparison. Subsequently, we have carried-out steady-state and time-resolved anisotropy measurements within these systems.

3.11. Steady-state and time-resolved fluorescence anisotropy measurements

Fluorescence anisotropy can provide insight into the local viscosity of any solubilizing media surrounding an anisotropy probe since fluorophore mobility is highly sensitive to the microviscosity, and it ultimately affects the loss of polarization of the probe experienced during its excited-state lifetime after excitation using polarized light. We have first measured the steady-state anisotropy of one of the common and popular such probes, perylene, to investigate the effect of MEA addition to the (Thy:DA) DES with varying MEA mole fractions from 0 to 0.50 at temperatures ranging from 293 to 343 K. Steady-state anisotropy $\langle r \rangle$ is estimated according to the expression:[38]

$$\langle r \rangle = \frac{I_{VV} + GI_{VH}}{I_{VV} + GI_{VH}} \quad (10)$$

where I_{VH} and I_{VV} are the horizontal and vertical components, respectively, of the fluorescence emission intensities measured when the sample is excited with vertically polarised light. The grating factor, G, is specific to any optical system and it corrects the differences between the emitted horizontal and vertical components of fluorescence intensity as a function of wavelength. Perylene $\langle r \rangle$ at different MEA mole fractions and temperatures are presented in Table S8 (ESI†). Understandably, $\langle r \rangle$ is observed to have its largest value in the DES sample with the highest dynamic viscosity (*i.e.*, for the sample with χ_{MEA} = 0.50 at 283 K) and its lowest value in the sample with the least dynamic viscosity [*i.e.*, in neat (Thy:DA) DES at 343 K]. This primarily implies that $\langle r \rangle$, reflecting the microviscosity, varies similarly with the bulk dynamic viscosity. Similar to the response of probe BPP, when $\chi_{MEA} > 0.09$, MEA-added (Thy:DA) DES samples exhibit higher $\langle r \rangle$ than that in both the neat (Thy:DA) DES and in neat MEA, suggesting unusually high microviscosity of the cybotactic region milieu of the perylene probe within the MEA-added (Thy:DA) DES system. Furthermore, $\frac{1}{\langle r \rangle}$ is plotted against T/η to assess the conformation of the data in terms of the Perrin equation[34] (eqn (11)) assuming no variation in the molecular volume of the rotating probe regardless of the change in the MEA composition and temperature:

$$\frac{1}{\langle r \rangle} = \frac{1}{\langle r_0 \rangle} + \frac{\tau RT}{\langle r_0 \rangle \eta V} \quad (11)$$

where $\langle r_0 \rangle$ is the limiting anisotropy when no diffusion occurs, τ is the fluorescence lifetime, R is the gas constant, V is the molecular volume of the rotating entity and η is the local viscosity of the cybotactic region of the fluorophore.

Here, we have assumed the local viscosity to be the same as the bulk dynamic viscosity. A reasonable linear correlation is observed (Fig. 14, $R^2 > 0.98$) between $\frac{1}{\langle r \rangle}$ and T/η, which again strongly suggests bulk dynamic viscosity and the cybotactic region microviscosity to correlate well and that the solvodynamic molecular volume of the probe also almost remains

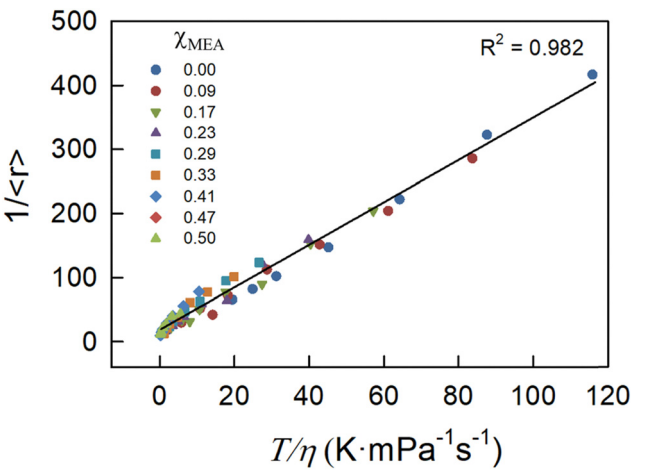


Fig. 14 $1/\langle r \rangle$ *versus* T/η for the MEA-added (Thy:DA) DES system. Solid line represents the outcome of the fit to the Perrin equation; $\frac{1}{\langle r \rangle} = \frac{1}{\langle r_0 \rangle} + \frac{\tau RT}{\langle r_0 \rangle \eta V}$.

 This journal is © the Owner Societies 2025

Published on 13 January 2025. Downloaded by Westlake University on 10/30/2025 1:55:24 AM.

constant with temperature and MEA composition variations. Excited-state anisotropy decay, $r(t)$, also affords insights into the local environment through the rotational diffusion of a rotor.[34] It not only depends upon the shape and size of the rotor; the cybotactic region surrounding the rotor molecule plays a major role as well. Two common and popular rotors, 1,6-diphenylhexatriene (DPH) and perylene, are employed to acquire the $r(t)$ within the investigated (Thy:DA) DES system varying the MEA mole ratio from 0 to 0.50 at 298 K. The collected $r(t)$ are fit to a simplistic sum of exponential decay expression:

$$r(t) = \sum r_{0,j} \exp(-t/\theta_j) \quad (12)$$

where $r_{0,j}$ is the limiting anisotropy when no rotational diffusion occurs and θ_j are the individual rotational reorientation times. Normally, rotational reorientation time (θ) depends upon the viscous drag offered by the solubilizing media. For an anisotropic rotor perylene, $r(t)$ is found to fit to a single exponential decay expression. We believe this to be due to faster in-plane rotation of the perylene probe, which is too rapid to be measured on our instrumental setup; hence, the observation of only out-of-plane rotation by us.

For the DPH rotor also, $r(t)$ decays single exponentially in the MEA-added (Thy:DA) DES system. The emission the dipole of DPH is oriented along its longer axis,[34] and the rotation that displaces this dipole is expected to be isotropic. In solvents, only a single rotational motion displaces the emission dipole of DPH resulting in single exponential decay of $r(t)$. The θ for perylene and DPH at the investigated MEA mole fractions are plotted in Fig. 15 and the values along with the corresponding goodness-of-fit in terms of χ^2 are listed in Table S9 (ESI†). As expected, θ of both the rotors increases rapidly with the increase in MEA mole fraction and experiences maximum hindrance in rotational motion at $\chi_{MEA} \sim 0.50$ (for perylene, θ = 3.60 ns; for DPH, θ = 14.5 ns). These are significantly higher than the θ in either neat (Thy:DA) DES (0.38 ns and 0.94 ns for perylene and DPH, respectively) or in neat MEA (0.73 ns and 2.61 ns for perylene and DPH, respectively) exhibiting minimum hindrance encountered by both the solutes in the neat DES. It is important to note that the rapid increase in dynamic viscosity is also manifested through the rotational diffusion dynamics of the solutes. Irrespective of the molecular structure, the solute cybotactic region appears to mimic the bulk medium within the MEA-added (Thy:DA) DES system in the context of affording hindrance to rotational dynamics. Here also, the bulk viscosity and microviscosity correlate well and the rotor experiences more hindrance in the MEA-added (Thy:DA) DES system compared to the neat DES and neat MEA (inset of Fig. 15).

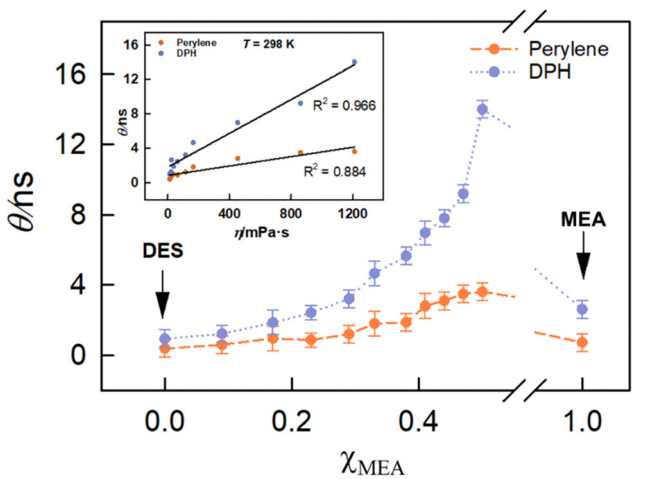


Fig. 15 Variation of rotational reorientation times (θ) of the rotors [perylene (1 μM), λ_{ex} = 405 nm and λ_{em} = 442 nm, DPH (1 μM), λ_{ex} = 308 nm and λ_{em} = 428 nm] dissolved in the MEA-added (Thy:DA) DES system at investigated mole fractions of MEA (χ_{MEA}) at 298 K. Dotted lines are just to guide the eyes. The inset of the graph shows the variation of rotational reorientation times (θ) with dynamic viscosity (η). Solid lines represent outcomes of the linear regression analysis.

4. Discussion

The anomalously high viscosity of terpenoid containing DA-based DESs upon addition of MEA is the most interesting aspect of this work considering no gel-phase is formed as clearly implied by the rheology measurements. The viscosity of the MEA-added DA-based DES systems is observed to even surpass the viscosity of the neat components – DES and MEA – at all the investigated compositions and temperatures, which is a rarely observed phenomenon. To the best of our knowledge, this is the first report documenting such viscosity 'synergism' involving DES systems. Similar behavior has primarily been observed in a few non-DES systems previously. For instance, viscosity synergism has been reported for ionic liquids mixed with polymers, such as [emim][OTf] + PEG MMA (6.2 ethoxy units in PEG) and [emim][BF_4] + PEG DMA (9 ethoxy units in PEG) at 40 °C, as well as for [bmim][PF_6] + PEG systems (PEG 200, 400, 600, and 1000) at specific temperatures and compositions.[39] Viscosity synergism in aqueous mixtures is reported for short-chain alcohols,[40] dimethyl sulfoxide,[41] hydroxypropyl-substituted guar and carboxymethyl hydroxypropyl-substituted guar,[42] acetone,[43] and isobutyric acid,[44] while in non-aqueous systems, it has been observed in only a few systems, such as (methanol + chloroform),[45] (*o*-chlorophenol + dioxane),[46] (carbon tetrachloride + ethyl acMEAte),[47] (dimethyl sulfoxide + acetic acid),[46] and (methanol + toluene).[47] The rarity of viscosity synergism within non-aqueous systems underscores the novelty of the current findings.

There are circumstantial pieces of evidence suggesting that even a slight modification in HBD and HBA constituents of DES can result in drastic alteration in their physicochemical properties, and hence, their interactions with other moieties.[6,19,48] In the present investigation, drastic alterations are induced by MEA as an additive that significantly perturbs the existing H-bonding network of the DES. Among the major alterations, a pronounced synergistic viscosity effect is observed in long chain fatty acid (*i.e.*, DA) based DESs in contrast to those DESs that do not have carboxylic acid as a constituent. We primarily

Published on 13 January 2025. Downloaded by Westlake University on 10/30/2025 1:55:24 AM.

hypothesize that an extensive H-bonding between DES and MEA involving the carboxylic acid functionality of DA could be one of the possible reasons for the observed synergistic effect in all the investigated systems. This hypothesis is duly supported by the observation that $(\ln \eta)^{E}$ decreases as temperature increases. Typically, this trend is observed in H-bonded systems where H-bond strength weakens as the temperature rises. FTIR absorbance data also supports this hypothesis suggesting that extended H-bonding is responsible for the synergistic effect in viscosity.

However, based on electrical conductivity and UV-vis absorbance data, it appears that in MEA-added DA-based DESs, the synergistic viscosity effect is not solely due to the formation of an extended H-bonding network, additional factors, such as induced ionic interactions, also contribute significantly in bringing out synergism. Due to the confluence of various factors, the system exhibits a divergence point ($\chi_{MEA} \sim 0.3$) beyond which viscosity and electrical conductivity experiences an exponential rise. FTIR absorbance has been highly instrumental in determining the extent of chemical and physical interactions present within the systems; for example, MEA-added (Thy:DA) has much higher chemical activity as compared to MEA-added (Thy:Men). Specifically, physical interactions are evident by subtle changes in the bending and stretching frequencies of various functional groups, such as, C=C, C–O, NH_2, –OH, *etc.*, upon addition of MEA. This implies the presence of strong physical interactions, H-bonding in particular. Chemical interactions are strongly detected in the stretching frequency of the carbonyl peak of DA along with the electrical conductivity and UV-vis absorbance data. Additionally, formation of new fluxional bonds or entities is further supported by the strong exothermic events at lower temperatures in the DSC profile of MEA-added DES systems, further supporting the presence of chemical interactions. Interestingly, ratio of excimer-to-monomer emission intensity of BPP and steady-state (perylene) and time-resolved (perylene and DPH) anisotropy measurements suggest that chemically-driven bulk viscosity mirror images the local cybotactic region viscosity (*i.e.*, microviscosity) sensed by the solute probes. The MEA-added (Thy:DA) DES systems exhibit both physicochemical changes and induced ionic interactions that result in higher and unexpected viscosity in comparison to the MEA-added (Thy:Men) DES system, which demonstrates only physical interactions albeit weaker in comparison.

5. Conclusion

The H-bonding and other interactions within DESs constituted of carboxylic acid as a constituent can be effectively manipulated and altered by addition of a judiciously selected external species. MEA, capable of readily forming H-bonding, when added to (Thy/Men:DA) DESs not only significantly alters the H-bonding network, but also creates induced ionic interactions within the DES system. These major microstructural changes amply get manifested in an anomalous increase in the viscosity and electrical conductivity of the system – both these properties attain unusually high values that are much higher than that of the neat DES and neat MEA. The MEA-added DA-based DESs behave as Newtonian fluids thus discarding the possibility of gel formation within the system. The appearance of color upon MEA addition to DA-based DESs hints at the possible presence of larger aggregates due to H-bonding and induced ionic interactions, which is further supported by significant alterations in the IR absorbance bands and the emergence of new bands perhaps due to the formation of fluxional bonds. Solute cybotactic region viscosity (microviscosity) mirrors the bulk dynamic viscosity, further supporting the above reasons for the physicochemical changes within the MEA-added DA-based DES systems. The work put-forth an effective way of modulating the properties of DESs by addition of judiciously selected additives without altering the DES constituents and composition. The outcome widens the application potential of DESs and further enhances their utility as new-age environmentally-benign inexpensive bio-derived solubilizing media.

Data availability

All the data that support the findings of this study are included within the paper.

Conflicts of interest

The authors declare no competing financial interest.

Acknowledgements

This work is generously supported by the Council of Scientific & Industrial Research, EMR-II (CSIR-EMR-II), Government of India through a grant to Siddharth Pandey [grant number 01(3043)/21/EMR-II]. We thank Science and Engineering Research Board (SERB), Government of India for funding [grant no. CRG/2021/000602] for the UV-Vis Spectrophotometer. We also thank the central research facility (CRF) of IIT Delhi, for providing access to the DSC facility.

References

1 N. V. Plechkova and K. R. Seddon, *Chem. Soc. Rev.*, 2007, **37**, 123–150.
2 F. Chemat, M. A. Vian, H. K. Ravi, B. Khadhraoui, S. Hilali, S. Perino and A. S. F. Tixier, *Molecules*, 2019, **24**(16), 3007.
3 J. Cao and E. Su, *J. Cleaner Prod.*, 2021, **314**, 127965.
4 M. Devi, R. Moral, S. Thakuria, A. Mitra and S. Paul, *ACS Omega*, 2023, **8**(11), 9702–9728.
5 J. Saien, M. Bahiraei and F. A. Jafari, *Sci. Rep.*, 2023, **13**(1), 1–11.
6 B. B. Hansen, S. Spittle, B. Chen, D. Poe, Y. Zhang, J. M. Klein, J. M. A. Horton, L. Adhikari, T. Zelovich, B. W. Doherty, B. Gurkan, E. J. Maginn, A. Ragauskas, M. Dadmun, T. A. Zawodzinski, G. A. Baker, M. E. Tuckerman,

 This journal is © the Owner Societies 2025

Published on 13 January 2025. Downloaded by Westlake University on 10/30/2025 1:55:24 AM.

R. F. Savinell and J. R. Sangoro, *Chem. Rev.*, 2021, **121**(3), 1232–1285.
7 E. L. Smith, A. P. Abbott and K. S. Ryder, *Chem. Rev.*, 2014, **114**(21), 11060–11082.
8 A. T. N. Fajar, T. Hanada, A. D. Hartono and M. Goto, *Commun. Chem.*, 2024, **7**(1), 1–10.
9 N. M. Stephens and E. A. Smith, *Langmuir*, 2022, **38**(46), 14017–14024.
10 N. F. Gajardo-Parra, L. Meneses, A. R. C. Duarte, A. Paiva and C. Held, *ACS Sustainable Chem. Eng.*, 2022, **10**(38), 12873–12881.
11 H. J. T. Carin, E. Annika, C. K. Maaike, V. S. A. Martin, G. Fausto and H. Christoph, *Fluid Phase Equilib.*, 2019, **489**, 75–82.
12 E. Cea-Klapp, N. F. Gajardo-Parra, P. Aravena, H. Quinteros-Lama, C. Held, R. I. Canales and J. M. Garrido, *Ind. Eng. Chem. Res.*, 2022, **61**, 2580–2591.
13 A. Paulo, C. K. Esteban, F. Nicolás, N. F. Gajardo-Parra, H. Christoph, M. G. José Matías and I. C. Roberto, *J. Mol. Liq.*, 2023, **389**, 122856.
14 H. Kivelä, M. Salomäki, P. Vainikka, E. Mäkilä, F. Poletti, S. Ruggeri, F. Terzi and J. Lukkari, *J. Phys. Chem. B*, 2022, **126**(2), 513–527.
15 S. Rozas, C. Benito, R. Alcalde, M. Atilhan and S. Aparicio, *J. Mol. Liq.*, 2021, **344**, 117717.
16 O. S. Hammond, D. T. Bowron and K. J. Edler, *Angew. Chem., Int. Ed.*, 2017, **56**(33), 9782–9785.
17 S. Bhattacharjee, R. Getman, B. Gurkan and R. Dikki, *J. Mol. Liq.*, 2024, **410**, 125569.
18 M. M. Nolasco, S. N. Pedro, C. Vilela, P. D. Vaz, P. Ribeiro-Claro, S. Rudić, S. F. Parker, C. S. R. Freire, M. G. Freire and A. J. D. Silvestre, *Front. Phys.*, 2022, **10**, 834571.
19 K. A. Omar and R. Sadeghi, *J. Mol. Liq.*, 2022, **360**, 119524.
20 A. Anjali and S. Pandey, *Langmuir*, 2024, **40**(4), 2254–2267.
21 L. Grunberg and A. H. Nissan, *Nature*, 1949, **164**(4175), 799–800.
22 S. L. Oswal and H. S. Desai, *Fluid Phase Equilib.*, 1998, **149**(1–2), 359–376.
23 A. Heintz, D. Klasen and J. K. Lehmann, *J. Solution Chem.*, 2002, **31**(6), 467–476.
24 E. N. D. C. Andrade, *Nature*, 1930, **125**(3148), 309–310.
25 R. K. Wadi and P. Saxena, *Indian J. Chem.*, 1995, **34**, 273–277.
26 M. K. Rajput, M. Konwar and D. Sarma, *Environ. Technol. Innovation*, 2021, **24**, 102017.
27 A. C. S. Valderrama and G. C. R. De, *Am. J. Anal. Chem.*, 2017, **8**(11), 726–741.
28 N. Fattahi, R. Heidari, B. Ghazanfaripoor, E. Masoudipour, J. Gharehdaghi and K. S. Nejad, *J. Pharm. Biomed. Anal.*, 2024, **238**, 115811.
29 F. A. Al-Bayati, *Ann. Clin. Microbiol. Antimicrob.*, 2009, **8**(1), 1–6.
30 D. L. Pavia, G. M. Lampman, G. S. Kriz and J. L. Vyvyan, *Introduction to Spectroscopy*, 5th edn, Cengage Learning, 2014.
31 M. J. Kamlet, J. L. Abboud and R. W. Taft, *J. Am. Chem. Soc.*, 1977, **99**(18), 6027–6038.
32 M. A. Ab Rani, N. Borduas, V. Colquhoun, R. Hanrefey, H. Johnson, S. Larger, P. D. Lickiss, V. Llopis-Mestre, S. Luu, M. Mogstad, P. Oczipka, J. R. Sherwood, T. Welton and J. Y. Xing, *Green Chem.*, 2014, **16**(3), 1282–1296.
33 M. J. Kamlet and R. W. Taft, *J. Am. Chem. Soc.*, 1976, **98**(2), 377–383.
34 A. S. Labban and Y. Marcus, *J. Chem. Soc., Faraday Trans.*, 1997, **93**(1), 77–79.
35 A. Kadyan and S. Pandey, *J. Phys. Chem. B*, 2017, **121**(5), 1081–1091.
36 S. Juneja, N. Himanshu and S. Pandey, *Phys. Chem. Chem. Phys.*, 2022, **24**(14), 8361–8370.
37 W. E. Acree, D. C. Wilkins, S. A. Tucker, J. M. Griffin and J. R. Powell, *J. Phys. Chem.*, 1994, **98**(10), 2537–2544.
38 J. R. Lakowicz, *Principles of Fluorescence Spectroscopy* 2006, pp. 1–954.
39 E. Andrzejewska, M. Podgorska-Golubska, I. Stepniak and M. Andrzejewski, *Polymer*, 2009, **50**(9), 2040–2047.
40 M. Ageno and C. Frontali, *Proc. Natl. Acad. Sci. U. S. A.*, 1967, **57**(4), 856–860.
41 S. A. Schichman and R. L. Amey, *J. Phys. Chem.*, 1971, **75**(1), 98–102.
42 L. M. Zhang and J. F. Zhou, *Colloids Surf., A*, 2006, **279**(1–3), 34–39.
43 S. Song and C. Peng, *J. Dispersion Sci. Technol.*, 2008, **29**(10), 1367–1372.
44 N. Ouerfelli, T. Kouissi, N. Zrelli and M. Bouanz, *J. Solution Chem.*, 2009, **38**(8), 983–1004.
45 A. M. Crabtree and J. F. O'Brien, *J. Chem. Eng. Data*, 1991, **36**(2), 140–142.
46 R. J. Fort and W. R. Moore, *Trans. Faraday Soc.*, 1965, **61**, 1112.
47 L. Qunfang and H. Yu-Chun, *Fluid Phase Equilib.*, 1999, **154**(1), 153–163.
48 T. El Achkar, H. Greige-Gerges and S. Fourmentin, *Environ. Chem. Lett.*, 2021, **19**(4), 3397–3408.